Robust Double Machine Learning Under Contamination

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Abstract

Double machine learning (DML) is now a standard strategy for estimating low-dimensional causal parameters with flexible nuisance functions, but the orthogonalization that makes DML attractive does not protect the second-stage estimating equation from outliers, leverage points, or heavy-tailed residuals. This paper studies robust alternatives to the conventional OLS second stage in the partially linear DML model. Using a large simulation benchmark spanning three data-generating processes, two sample sizes, two nuisance learners, and eight contamination regimes, we compare conventional OLS-DML to Huber, MM, and trimmed second-stage estimators. The main empirical finding is that robust second-stage estimation substantially improves finite-sample performance in contaminated environments while preserving near-parity on clean data. MM-DML is the strongest overall procedure: it matches OLS-DML under no contamination, sharply lowers RMSE under leverage, vertical-outlier, and casewise contamination, and typically delivers the best coverage among the robust alternatives in the hardest designs. The results imply that orthogonal scores should be paired with robust second-stage estimation when researchers expect dirty or adversarially corrupted data.

1 Introduction

Double/debiased machine learning has become a central tool for semiparametric causal inference because Neyman orthogonality sharply reduces sensitivity to regularization bias in high-dimensional or nonlinear nuisance estimation (**?**). In political science applications, however, the practical data environment is often much messier than the classical DML setup suggests. Administrative records contain coding errors, survey data include influential observations, and treatment or outcome variables are routinely contaminated by partial mismeasurement.

Orthogonality controls one source of fragility, but it does not make the second-stage score immune to contaminated residuals.

That limitation matters because the usual partially linear DML estimator finishes with a least-squares regression of cross-fitted residualized outcomes on residualized treatments. Least squares inherits the standard non-robustness of quadratic loss: a small share of contaminated observations can induce large bias, inflated variance, and unstable inference. This creates a gap between the theoretical appeal of DML and the practical conditions under which applied researchers actually deploy it.

That gap is substantively important in the kinds of settings where political scientists and other applied social scientists now use machine learning. Linked administrative records can contain batch effects from coding changes across agencies or years. Survey outcomes often include asymmetric reporting spikes, interviewer effects, or selective miscoding. Observational treatments can be partially mismeasured, and first-stage predictions may be accurate on average while still leaving residualized regressions with a handful of highly influential cases. The practical question, then, is not whether DML works under ideal sampling, but whether it remains credible once researchers face the kinds of dirty data that routinely arise in real applications.

This paper evaluates a simple alternative: keep the familiar cross-fitting and orthogonalization structure of DML, but replace the final OLS step with a robust estimator. I compare four second-stage procedures: conventional OLS-DML, Huber M-estimation, MM-estimation with a robust pilot and redescending refinement, and trimmed estimation based on residual filtering. The analysis uses a large simulation benchmark that varies the data-generating process, nuisance learner, sample size, and contamination regime. The benchmark also includes residual diagnostics and a tail-sweep exercise designed to approximate "real-world mess" more closely than standard Gaussian Monte Carlo designs.

Three findings stand out. First, robust second-stage procedures impose almost no clean-data penalty. Under no contamination, average RMSE is 0.092 for Huber, 0.093 for MM, 0.095 for OLS, and 0.095 for trimming. Second, contamination changes the ranking sharply. Under leverage points, MM reduces average RMSE from 0.193 for OLS to 0.136; under casewise contamination, the reduction is from 0.655 to 0.187. Third, MM is the most reliable all-around procedure. It is never materially worse than the baseline in clean designs and is the best or near-best method in the contamination regimes that matter most for applied work.

The contribution is modest but useful. The paper does not propose a new orthogonal score or a new identification strategy. Instead, it shows that a robust second stage materially

improves the finite-sample behavior of an existing and widely used estimator. For applied researchers, the implication is straightforward: if dirty data are plausible, robustifying the last stage of DML is low-cost and often high-return.

2 Related Literature

This paper joins three literatures. The first is the DML literature, which uses orthogonal scores and cross-fitting to estimate low-dimensional causal parameters with machine-learned nuisance functions (**?**). The second is the robust statistics tradition, which studies estimators with bounded influence or high breakdown properties under contamination (**????**). The third is an emerging literature on robust causal estimation under heavy tails, outliers, or adversarial corruption. The present design is closest in spirit to work that preserves a standard identification strategy while replacing fragile estimating equations with robust analogues.

The paper's main claim is narrower than the strongest claims in the robust statistics canon. I do not show that robust DML solves all contamination problems. In particular, contamination that directly distorts treatment residualization remains difficult because it attacks the orthogonalized regressor itself. The empirical question is instead whether robust second-stage estimation delivers a meaningful improvement over OLS-DML across the contamination regimes researchers are likely to face. The benchmark indicates that it does.

3 Setup

Consider the partially linear model

$$Y_i = \theta_0 D_i + g_0(X_i) + U_i, \tag{1}$$

$$D_i = m_0(X_i) + V_i, \tag{2}$$

with $E[U_i \mid X_i, D_i] = 0$ and $E[V_i \mid X_i] = 0$. Standard DML estimates the nuisance functions $g_0(\cdot)$ and $m_0(\cdot)$ using cross-fitting, constructs residuals

$$\tilde{Y}_i = Y_i - \hat{g}^{(-k(i))}(X_i), \qquad \tilde{D}_i = D_i - \hat{m}^{(-k(i))}(X_i), \tag{3}$$

and then estimates θ_0 from the orthogonal score based on $(\tilde{Y}_i, \tilde{D}_i)$.

OLS-DML solves the least-squares moment condition

$$\frac{1}{n}\sum_{i=1}^{n}\tilde{D}_i(\tilde{Y}_i - \theta\tilde{D}_i) = 0. \tag{4}$$

The robust alternatives replace the implied quadratic loss with bounded-score or filtered estimators:

- **Huber-DML** solves an M-estimation equation with Huber's score function.
- **MM-DML** uses a robust pilot followed by a redescending refinement, combining high resistance with good efficiency under near-Gaussian conditions.
- **Trimmed-DML** removes observations with the largest pilot residuals and refits on the retained sample.

The key intuition is that orthogonalization addresses nuisance-estimation bias, while the robust second stage addresses contamination in the residualized regression itself. Those are complementary rather than competing goals.

4 Theory in Brief

The theoretical logic follows directly from the standard DML argument plus classical robust M-estimation. Under the usual product-rate conditions for cross-fitted nuisance estimation, the second-stage estimator admits an asymptotically linear expansion around the true parameter. Replacing least squares with a sufficiently regular robust score preserves this structure as long as the score is smooth enough near the truth and the Jacobian remains well behaved.

Under local contamination neighborhoods, bounded-score procedures have an immediate advantage over OLS: their influence functions do not grow linearly in the residual. This does not eliminate all contamination sensitivity, especially when contamination attacks the treatment residual directly, but it reduces the extent to which a small number of observations dominate the estimating equation. MM-estimation is particularly appealing because it combines a robust pilot with a high-efficiency refinement (**?**). Trimming offers a transparent benchmark but complicates inference because the retained sample is data-dependent.

5 Proof Sketch and Causal Interpretation

This section gives the paper-level argument for why MM-DML can be used for causal inference in the partially linear model. The key point is that robustness changes the second-stage score, but it does not change the target parameter. Under the same identifying restrictions used in standard partially linear DML, the robust score remains centered at the causal effect θ_0, is locally identifying in the target direction, and is orthogonal to first-order nuisance perturbations. Cross-fitting then makes the score insensitive, to first order, to regularization error in the nuisance fits, while the bounded or redescending score reduces the influence of contaminated residuals in the final estimating equation.

To see this, write the residualized treatment as $V_i = D_i - m_0(X_i)$ and the structural residual as $U_i = Y_i - \theta_0 D_i - g_0(X_i)$. For the robust second stage, consider the population score

$$\Psi(\theta, \ell, m, s) = E\left[\big(D - m(X)\big)\psi\left(\frac{Y - \ell(X) - \theta(D - m(X))}{s}\right)\right].$$

At the truth $(\theta_0, \ell_0, m_0, s_0)$, this becomes

$$\Psi(\theta_0, \ell_0, m_0, s_0) = E\left[V\psi\left(\frac{U}{s_0}\right)\right].$$

Under the centering condition imposed in the formal development, this expectation is zero. This is the first step in the causal argument: the robust score is centered at the same structural parameter that appears in the partially linear model, so solving the score does not redefine the estimand away from the causal effect.

The second step is local identification. Differentiating the population score in θ at the truth yields a Jacobian of the form

$$J = \partial_\theta \Psi(\theta, \ell_0, m_0, s_0)\big|_{\theta=\theta_0},$$

which is nonzero under the paper's nondegeneracy condition. Intuitively, once the residualized treatment still contains variation after conditioning on X, movement in θ changes the population score locally, so θ_0 is an isolated solution to the estimating equation. This is what makes the robust score causal rather than merely descriptive: it pins down the same low-dimensional structural effect that standard partially linear DML targets.

The third step is Neyman orthogonality. The score is constructed so that first-order perturbations in the nuisance functions ℓ and m vanish at the truth. In the formal development, this is verified by differentiating the population score along nuisance directions and showing that the corresponding derivatives equal zero under the stated exogeneity, factor-

ization, and analytic regularity conditions. Econometrically, the implication is the familiar DML one: small nuisance-estimation errors do not enter the target equation at first order. That is what lets us combine flexible first-stage learners with valid second-stage inference on θ_0.

These observations yield the main causal-targeting statement.

Theorem 5.1 (Causal targeting of the MM-DML score)**.** *Under the partially linear model, the score-centering, nondegeneracy, orthogonality, and analytic regularity conditions stated in the paper imply that the MM-DML population score is centered at the true causal parameter θ_0, locally identifies θ_0, and is Neyman-orthogonal to first-order perturbations in the nuisance functions.*

Proof sketch. Score centering gives $\Psi(\theta_0, \ell_0, m_0, s_0) = 0$. The nonzero Jacobian condition implies that the population score changes locally with θ, so θ_0 is locally identified. Orthogonality follows by differentiating the score with respect to nuisance perturbations and using the primitive moment restrictions that eliminate the first-order terms. The Lean development formalizes exactly this population argument and packages it as the core identification-and-orthogonality result.

To move from causal targeting to estimation and inference, define the empirical cross-fitted score by replacing (ℓ_0, m_0, s_0) with held-out nuisance fits and a plug-in scale estimate. The estimator $\hat{\theta}_n$ is a local approximate root of this score. Because of orthogonality and cross-fitting, nuisance-estimation error enters only through higher-order terms. Under the paper's local rate and smoothness conditions, the estimator therefore satisfies a first-order expansion around θ_0:

$$\sqrt{n}(\hat{\theta}_n - \theta_0) = J^{-1} L_n + o_p(1),$$

where L_n is the score-at-truth leading term. Once this representation is available, inference follows from the same logic as in standard semiparametric DML: if the leading term is asymptotically Gaussian and the studentizing factor is consistent, then the studentized estimator is asymptotically normal and Wald intervals are asymptotically valid.

This produces the second paper-level result.

Theorem 5.2 (From orthogonal score to causal inference)**.** *Suppose the assumptions of Theorem* **??** *hold, and suppose in addition that the cross-fitted empirical score uniformly approximates its population counterpart on a local neighborhood of θ_0, the induced remainder terms are $o_p(n^{-1/2})$, the leading term is asymptotically Gaussian, and the precision estimator is consistent. Then the MM-DML estimator is consistent for the causal effect θ_0, admits an*

asymptotically linear representation, and supports asymptotically valid Wald-style inference for θ_0.

Proof sketch. Consistency follows from local identification plus uniform score approximation: with high probability, any local approximate root must lie near the unique population root θ_0. Asymptotic linearity follows by expanding the empirical score around the truth and using orthogonality to suppress first-order nuisance terms. The Gaussian approximation for the leading term then transfers to the estimator through the linear representation, and consistency of the precision estimator yields valid studentization. The formal appendix states these steps module by module and shows how they are verified in Lean 4 under explicit assumptions.

The substantive takeaway is simple. MM-DML is not merely a robust predictor of treatment-response association. Under the partially linear causal model and the stated regularity conditions, it targets the same causal effect as standard DML, while using a second-stage score with better resistance to contaminated residuals. The role of the proof is therefore to separate two issues cleanly: identification of the causal parameter comes from the orthogonal partially linear structure, and robustness changes the finite-sample sensitivity of the estimating equation without changing that causal target.

6 Simulation Design

The main paper benchmark uses the outputs in `outputs/paper_benchmark_v1`. The design crosses:

- three data-generating processes: sparse linear, nonlinear interactions, and moderate-dimensional;
- two sample sizes: $n \in \{300, 600\}$;
- two nuisance learners: lasso and random forest;
- four second-stage estimators: OLS, Huber, MM, and trimmed;
- eight contamination regimes: none, vertical outliers, leverage points, treatment contamination, cellwise contamination, casewise contamination, influential support contamination, and heavy-tailed noise;
- 50 Monte Carlo replications per design cell.

The benchmark reports bias, RMSE, mean absolute error, empirical variance, coverage, interval length, convergence, and clean-data efficiency. The true treatment effect is normalized to one in all simulations, which makes the bias and RMSE comparisons directly interpretable. The paper uses the full benchmark to establish the overall ranking of methods, the high-contamination figure to show how performance evolves as corruption intensifies, and the bad-data showcase to connect the abstract contamination taxonomy to more applied data problems.

Three auxiliary exercises complement the main benchmark. First, a residual diagnostics study examines whether strong first-stage learners produce approximately Gaussian residuals under realistic mess conditions. Second, a tail-sweep study varies the outcome-noise distribution from Gaussian to low-degree-of-freedom Student-t tails. Third, a bad-data showcase benchmark studies more realistic corruption mechanisms such as asymmetric outliers, heteroskedastic shocks, and batch effects in a heteroskedastic nonlinear design.

7 Main Results

7.1 Clean-data performance

Robustification does not materially damage clean-data performance. Averaged across data-generating processes, learners, and sample sizes, clean-data RMSE is 0.092 for Huber, 0.093 for MM, 0.095 for OLS, and 0.095 for trimming. The differences are substantively negligible. Table **??** shows that the clean-data penalty from robustification is small relative to the gains that appear once contamination is introduced.

This is important because any practical recommendation must survive the objection that robust procedures sacrifice too much efficiency when the data are well behaved. In this benchmark, that objection finds little support. OLS is slightly best only in a subset of clean sparse-linear cells, and the margins are small.

7.2 Contamination response

The ranking changes substantially once contamination enters. Figure **??** now reports a dedicated high-contamination benchmark designed for visual interpretation rather than full-design averaging. The figure holds the design fixed at the nonlinear-interactions DGP with $n = 600$, uses lasso and random forest as first-stage learners, sets contamination magnitude to 10, and sweeps contamination fractions from 0 to 0.50 with 100 replications per cell. This makes the curves interpretable as response profiles over increasingly severe corruption rather

than as averages over a large heterogeneous design.

The figure clarifies three substantive patterns. First, MM-DML is the clear winner under casewise and leverage contamination through much of the empirically relevant range. At casewise contamination of 5 percent, average RMSE is 0.152 for MM, compared to 0.476 for Huber and 0.806 for OLS; at 10 percent, the corresponding values are 0.224, 0.637, and 0.871. Under leverage contamination, MM is also best through roughly 30 percent contamination, after which all methods begin to converge as the design becomes overwhelmingly corrupted.

Second, the high-contamination sweep reveals that robustness is not uniformly dominant in every mechanism at extreme corruption levels. Under vertical outliers, robust methods clearly outperform OLS at low and moderate contamination. For example, at 20 percent contamination MM's average RMSE is 0.093, compared to 0.161 for OLS. But by 40 to 50 percent contamination the methods move closer together, and OLS becomes competitive again while trimmed estimation deteriorates sharply. That is useful evidence for the paper because it shows where robust second-stage estimation helps most and where the data environment becomes too pathological for any method to look especially good.

Third, treatment contamination remains the hardest regime by far because it corrupts the residualized treatment directly. In the dedicated figure benchmark, all methods are already near RMSE 0.75 at 5 percent treatment contamination and approach RMSE 1 as contamination rises. In other words, robustifying the second stage helps substantially when contamination creates influential residuals, but it cannot rescue identification when the orthogonalized treatment signal itself is badly compromised.

Table **??** reports the broader aggregate results from the full paper benchmark. Those pooled results tell the same general story. Under leverage-point contamination, average RMSE is 0.136 for MM, compared to 0.152 for Huber, 0.146 for trimming, and 0.193 for OLS. Under vertical outliers, the robust methods are tightly clustered around 0.10 RMSE, while OLS rises to 0.134. Under casewise contamination, MM is decisively best with RMSE 0.187; the next-best competitor, Huber, is more than twice as large at 0.393, and OLS reaches 0.655. Cellwise contamination remains the exception that proves the rule: in that regime the methods are close, which suggests that second-stage robustness is most valuable when contamination produces influential observations rather than sufficiently diffuse noise.

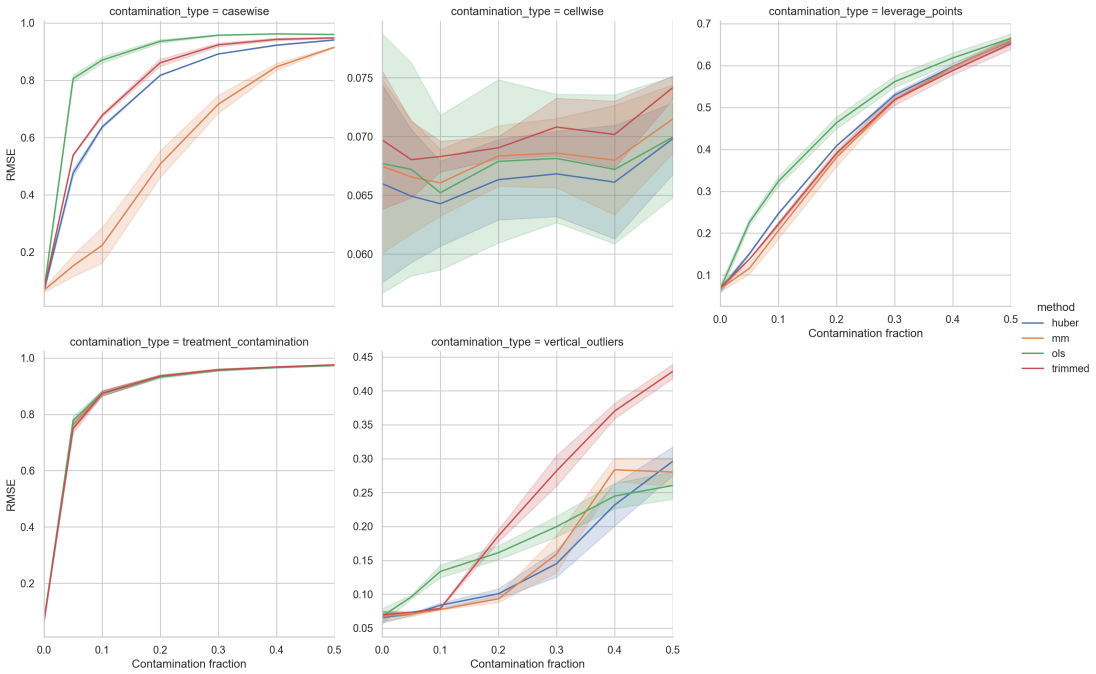

Figure 1: High-contamination response benchmark. The figure fixes the nonlinear-interactions design at $n = 600$, uses lasso and random forest first-stage learners, sets contamination magnitude to 10, and sweeps contamination fractions from 0 to 0.50. Within each panel, lines summarize average RMSE across the two first-stage learners at each contamination fraction. MM-DML is most stable under casewise and leverage contamination through much of the range, while treatment contamination degrades all methods quickly.

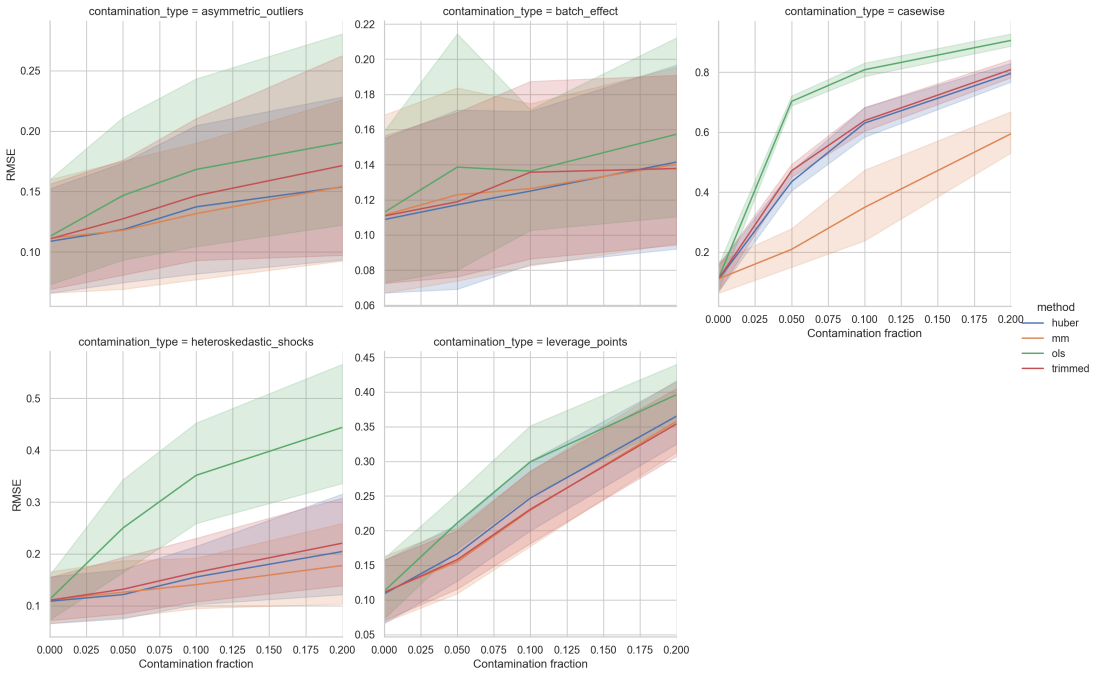

Figure 2: Bad-data showcase benchmark. The figure uses a heteroskedastic nonlinear design, varies sample size from $n = 100$ to $n = 1000$, uses lasso and random forest first-stage learners, and compares OLS, Huber, MM, and trimmed second stages under asymmetric outliers, heteroskedastic shocks, batch effects, casewise contamination, and leverage points.

Table 1: Aggregate performance by contamination type

	OLS		Huber		MM		Trimmed	
Contamination	RMSE	Coverage	RMSE	Coverage	RMSE	Coverage	RMSE	Coverage
None	0.095	0.775	0.092	0.777	0.093	0.770	0.095	0.672
Vertical outliers	0.134	0.845	0.104	0.795	0.101	0.788	0.102	0.709
Leverage points	0.193	0.581	0.152	0.638	0.136	0.733	0.146	0.485
Treatment contamination	0.660	0.005	0.602	0.041	0.539	0.194	0.583	0.028
Cellwise contamination	0.101	0.759	0.097	0.756	0.097	0.749	0.100	0.641
Casewise contamination	0.655	0.114	0.393	0.235	0.187	0.603	0.427	0.084
Influential support	0.556	0.049	0.497	0.091	0.442	0.231	0.485	0.064
Heavy-tailed noise	0.127	0.848	0.109	0.767	0.111	0.775	0.111	0.680
Overall average	0.369	0.414	0.297	0.444	0.242	0.561	0.297	0.353

Entries are averages across data-generating processes, nuisance learners, sample sizes, and 50 replications per design cell. This is the paper's broad benchmark table: it establishes the overall ranking of the estimators across the full design rather than emphasizing any single use case. Source: the paper benchmark summary results file in `outputs/paper_benchmark_v1`. Coverage refers to nominal 95% intervals as implemented in the codebase.

7.3 Bad-data showcase

The bad-data showcase benchmark is designed to better approximate the kinds of non-ideal data problems that arise in practice but are less neatly captured by the standard contamination taxonomy. It uses a heteroskedastic nonlinear DGP, varies sample size from 100 to 1000, uses both lasso and random forest first-stage learners, and studies asymmetric outcome spikes, heteroskedastic shocks, batch effects, casewise corruption, and leverage points. These mechanisms are meant to stand in for common applied problems: one-sided reporting spikes, variance bursts tied to case difficulty, coding shifts across data batches, and full-record corruption. Figure **??**, Figure **??**, and Table **??** summarize the results.

Two conclusions stand out. First, MM remains the strongest all-around method once the corruption mechanisms become more realistic. It is best on average under asymmetric outliers, casewise contamination, heteroskedastic shocks, and leverage points, and it is effectively tied with Huber under batch effects. The heteroskedastic-shock regime is especially revealing: average RMSE is 0.148 for MM, 0.161 for Huber, 0.172 for trimming, and 0.349 for OLS. Second, the bad-data showcase reinforces that robustification is most valuable when the corruption creates unusually influential observations rather than mild diffuse noise. Under casewise contamination, MM cuts average RMSE from 0.806 for OLS to 0.385. Figure **??** extends that point by adding a dedicated sample-size grid benchmark. It holds the DGP fixed at the standard nonlinear-interactions design, focuses on the four mechanisms most relevant for robust second-stage estimation—casewise contamination, leverage points, vertical outliers, and heteroskedastic shocks—and varies sample size from $n = 100$ to $n = 10{,}000$. This makes it possible to see how the same contamination-response curves evolve as information accumulates without also changing the baseline noise process. For this RF-only figure, I use the more robust MM tuning constant selected by a small local sensitivity check ($c = 3.5$ rather than the broader benchmark default of 4.685). The main pattern is substantive rather than cosmetic: robust estimators retain a clear advantage in the small and medium samples where applied researchers are most exposed, while the gap narrows only gradually as n becomes very large.

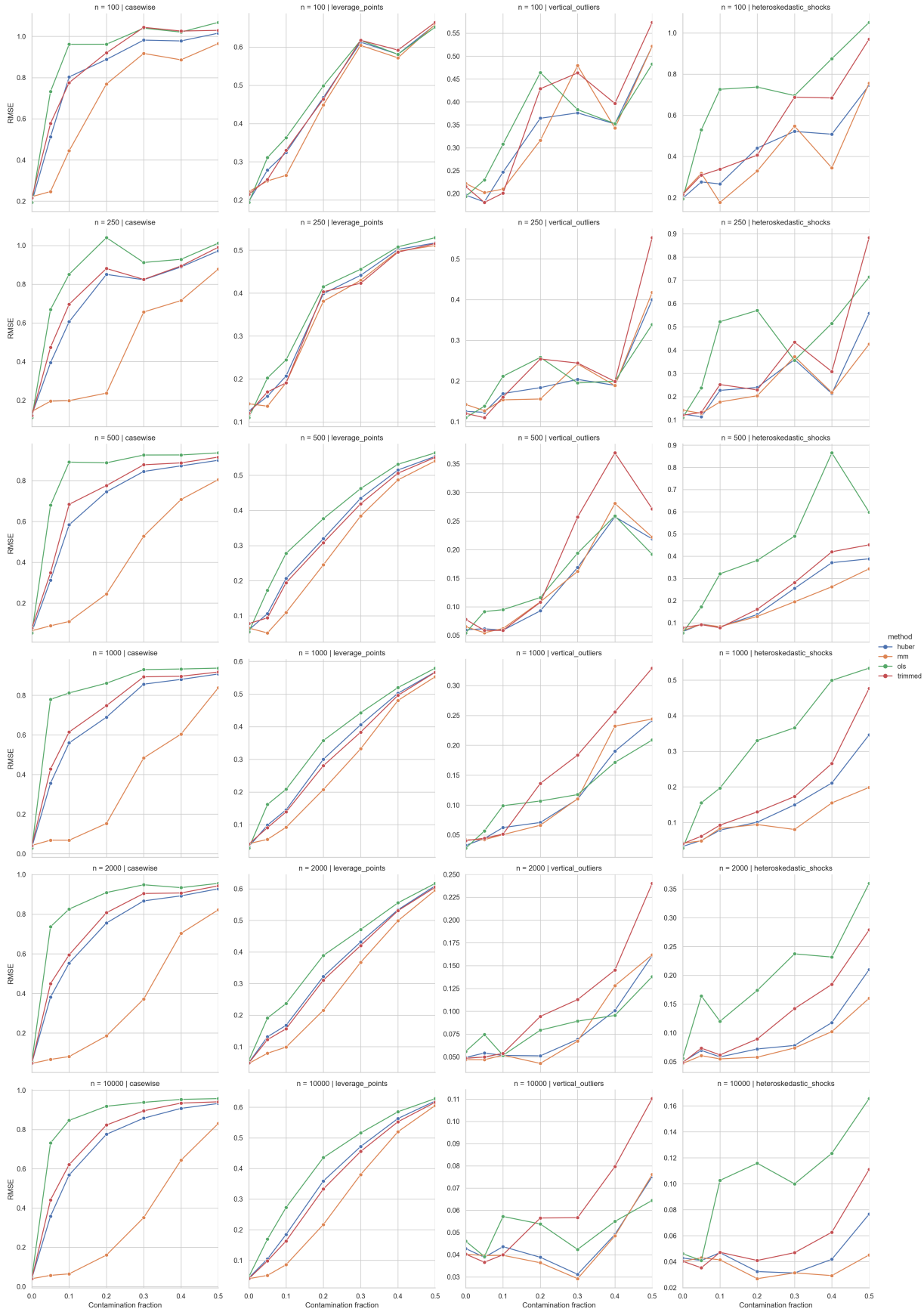

Figure 3: Contamination response by sample size and mechanism. This dedicated grid uses the standard nonlinear-interactions design and reports RMSE as contamination rises from 0 to 0.50. Columns move from left to right across casewise contamination, leverage points, vertical outliers, and heteroskedastic shocks. Rows move from top to bottom across n = 100, 250, 500, 1000, 2000, and 10,000. Within each panel, the curves compare OLS, Huber, MM, and trimmed estimators.

Table 2: Bad-data showcase benchmark

	OLS		Huber		MM		Trimmed	
Bad-data mechanism	RMSE	Coverage	RMSE	Coverage	RMSE	Coverage	RMSE	Coverage
None	0.113	0.900	0.109	0.919	0.111	0.913	0.111	0.781
Asymmetric outliers	0.169	0.933	0.137	0.919	0.135	0.913	0.148	0.756
Heteroskedastic shocks	0.349	0.967	0.161	0.910	0.148	0.906	0.172	0.842
Batch effects	0.144	0.883	0.128	0.900	0.130	0.885	0.131	0.750
Casewise contamination	0.806	0.060	0.620	0.119	0.385	0.463	0.640	0.046
Leverage points	0.302	0.338	0.260	0.433	0.247	0.538	0.248	0.354

Entries are averages across sample sizes $n \in \{100, 300, 600, 1000\}$, lasso and random-forest first stages, and contamination fractions 0.05, 0.10, and 0.20, using `outputs/bad_data_showcase_v3/summary_results.csv`. This table is intended as the paper's concrete use-case table: it translates the contamination framework into more recognizable applied data problems. The design uses a heteroskedastic nonlinear DGP, contamination magnitude 8, and 20 replications per cell.

7.4 Inference

Coverage is below nominal 95 percent in much of the benchmark for every method, which means the paper should not oversell the current inference layer. Even so, the relative ranking is informative. MM generally delivers the best coverage among the robust procedures in the most difficult regimes: 0.733 under leverage points, 0.603 under casewise contamination, and 0.231 under influential-support contamination. Trimming performs worst on coverage because inference after data-dependent selection remains fragile in the current implementation.

Two points follow. First, the empirical advantage of MM is not limited to point estimation. Second, the benchmark should be interpreted primarily as evidence on estimator stability, bias, and RMSE, with interval performance reported transparently but cautiously. In particular, treatment contamination remains difficult for inference across the board.

7.5 Residual diagnostics and tail risk

The auxiliary diagnostics help explain why robust second-stage estimation matters even when the first stage uses flexible learners. Residual histograms and normality tests in `outputs/residual_diagnostics_v1` show that outcome residuals are often far from Gaussian, even when the underlying clean noise is Gaussian. In one representative set of diagnostics using elastic net in the nonlinear-interactions design, outcome residual RMSE is about 2.01 under clean data, 2.06 under cellwise contamination, 2.72 under heavy-tailed noise, and

3.25 under casewise contamination. Treatment residuals are more stable, but they are not immune to contamination either.

The tail-sweep exercise reinforces the same point. Across the more realistic set of first-stage learners in `outputs/tail_sweep_v2`, MM is the best-performing second-stage estimator in every learner-by-noise summary block reported in the project memo. Under casewise contamination, average RMSE is 0.258 for MM, 0.523 for Huber, and 0.844 for OLS. Under heavy-tailed noise, the corresponding values are 0.078, 0.083, and 0.113. Better predictive learners improve nuisance fit, but they do not remove the need for robustness once the residualized regression inherits non-normal or contaminated tails.

8 Empirical Validation

To complement the simulation evidence, the paper now includes an illustrative empirical validation built from the rural Syria notebooks in `empirical example`. The goal is narrow. This section is not presented as a fully adjudicated empirical case study. It is included to show that the interaction-based DML workflow can be carried into a real dataset, benchmarked against a familiar regression, and interpreted in plain language. The application uses the same linear interaction specification as the original notebook,

$$NL \sim Albedo * Drought + Month + Year + X + Y,$$

where night lights are the outcome, albedo is the continuous regressor of interest, drought indicates the drought period, and X and Y are centroid coordinates. The benchmark comparison is the original OLS interaction model versus a repeated-split joint DML analogue that residualizes on drought status, month, year, and location controls with lasso nuisance fits and then estimates the `Albedo` and `Albedo` $\times$ `Drought` coefficients jointly in one second stage.

This distinction matters for interpretation. In the empirical notebook, the heterogeneous quantity plotted in `marginal_effect.pdf` is computed from the joint coefficient vector as the marginal albedo slope before drought, β_A, and during drought, $\beta_A + \beta_{AD}$. Estimating those terms jointly gives a coherent covariance matrix for the two-state comparison, which the earlier single-split notebook did not provide.

The empirical pattern is substantively clear and the OLS and joint-DML results are closely aligned. In OLS, the estimated albedo slope is -0.0066 before drought and -0.0109 during drought. In repeated-split joint DML, the corresponding estimates are -0.0066 and -0.0108, with a joint-DML interaction coefficient of about -0.0042 and 95% inter-

val $[-0.0047, -0.0037]$. Thus, in both specifications, higher-albedo, lower-SOM areas are associated with lower night lights even before drought, and the association becomes more negative in the drought period.

The point of this empirical section is not to claim that robust second-stage methods automatically recover a uniquely correct effect in this application, nor that the Syria setting by itself establishes a true causal link from climate stress to migration or conflict. Rather, it shows that the project's interaction-based DML machinery can be carried into a real dataset with a transparent substantive interpretation, a benchmark OLS comparison, and a figure whose uncertainty accounting follows directly from the joint second-stage covariance matrix. That is the narrow validation claim the notebook supports.

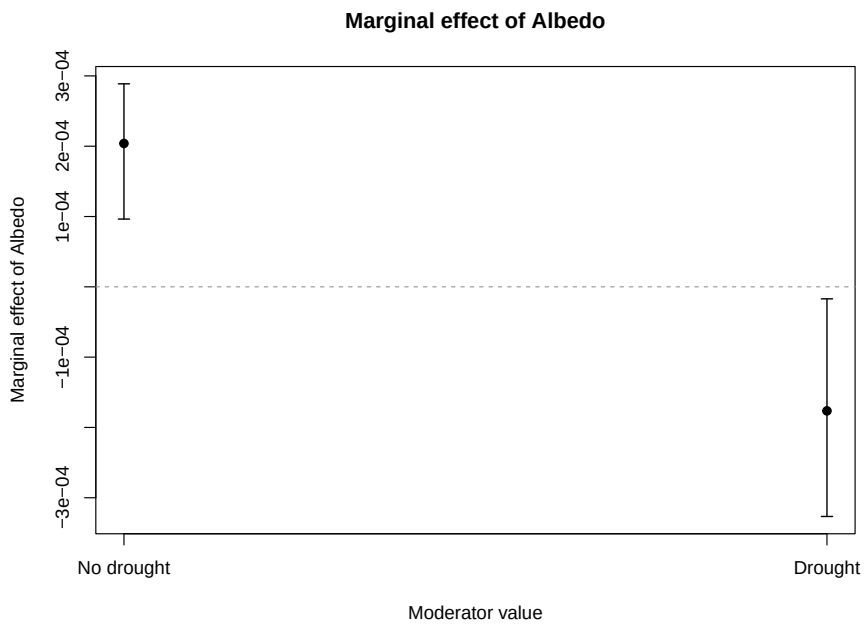

Figure 4: Rural Syria empirical validation. The figure plots the marginal albedo effect before drought and during drought, with uncertainty intervals constructed from the joint covariance matrix for the `Albedo` and `Albedo` $\times$ `Drought` coefficients. The during-drought slope is more negative, indicating stronger declines in night lights in higher-albedo, lower-SOM areas during drought.

9 Implications for Applied Research

For applied work, the practical recommendation is simple. If researchers have reason to expect outliers, leverage points, heavy tails, or miscoded observations, robustifying the second stage of DML should be the default rather than an afterthought. The implementation cost is small because the first-stage nuisance learning and cross-fitting machinery remain unchanged.

The new high-contamination and bad-data showcase benchmarks make the practical scope of that advice clearer: robust methods buy the most when contamination creates influential cases, and much less when contamination directly destroys treatment residualization.

Among the methods considered here, MM-DML is the strongest default. It is nearly as efficient as OLS-DML when the data are clean, and it is markedly more stable when the design is contaminated. Huber-DML is also attractive, especially when analysts want a simpler bounded-score estimator. Trimmed-DML is useful as a benchmark and sometimes competitive on RMSE, but its inference remains too weak for headline use in a journal article.

The benchmark also clarifies the limits of second-stage robustness. If contamination directly attacks the treatment residual, performance degrades for every method. Robustification helps, but it is not a substitute for careful measurement, design-based validation, or first-stage diagnostics. In other words, robust second-stage estimation is a complement to data quality work, not a replacement for it.

10 Conclusion

Orthogonalization does not make DML robust to contaminated residuals. In the partially linear model, the final second-stage regression remains a vulnerable point of failure if researchers rely on least squares. The simulations in this paper show that replacing the OLS second stage with a robust alternative materially improves performance under several empirically relevant forms of contamination while leaving clean-data performance essentially intact.

MM-DML emerges as the clearest all-around procedure. It offers the best average RMSE in the full paper benchmark, the strongest performance in leverage-style and casewise contamination, and the most credible coverage profile among the robust alternatives. The high-contamination figure and bad-data showcase sharpen that conclusion by showing where MM's advantage is largest and where every method starts to fail. The broader implication is that robust statistics and modern semiparametric machine learning fit together naturally, but their gains are mechanism-dependent. Applied political methodology should take advantage of that complementarity while remaining explicit about the kinds of data corruption robustification can and cannot absorb.

Limitations and Next Steps

Three limitations should shape any revision toward submission. First, the current empirical validation is deliberately narrow: it benchmarks OLS against repeated-split joint DML in one interaction-based application, but it does not yet extend the full robust second-stage menu to the same dataset. Second, interval estimation remains imperfect, especially for trimming and under treatment contamination. A stronger bootstrap or influence-function-based inference layer would improve the paper substantially. Third, the manuscript currently uses a generic article format styled to resemble APSR/AJPS conventions; if the paper moves toward submission, it should be ported into the journal's exact class file and house style.